

RAVINDRAN S

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AI/ML Engineer and Arch Linux tester with expertise in computer vision, edge AI, deep learning, and scalable distributed systems.
Consistent hackathon winner with a track record of delivering production-grade models and full-stack AI applications.

EXPERIENCE

- AI/ML Virtual Intern | AICTE and EduSkills** Apr 2025 – Jun 2025
- Streamlined the model inference pipeline through optimized data preprocessing, reducing end-to-end response latency by approximately 30%.
- Object Detection Engineer | CIT Drone Technology Team (NIDAR)** Oct 2025 – Dec 2025
- Designed and trained a sub-1-GFLOP convolutional neural network for drone-based object detection, attaining 96% precision while meeting strict edge-deployment constraints.
- Open Source Contributor | AWS (s2n-tls) and SWOC** 2024 – Present
- Strengthened the AWS s2n-tls TLS library by contributing code patches and documentation; resolved diverse pull requests during SWOC, cutting project-wide defects by approximately 90%.

PROJECTS

- MicroDet | TinyGrad, PyTorch, TensorRT, DeepStream SDK**
- Engineered a lightweight object-detection model (under 1 GFLOP) using a NanoDet backbone with YOLO-style heads, achieving 96% mean average precision with real-time on-device inference. Presented at NIDAR 2025.
- Machine Guard | Kotlin, Python, MQTT, Firebase, IoT Sensors**
- Built an Internet of Things (IoT) system for live industrial machine and environmental monitoring with 94% classification accuracy; official submission and winner (INR 50,000) at Ti Forge 2026.
- GenuineGate | Go, Redis, Nginx, JavaScript | github.com/ravindran-dev/GenuineGate**
- Developed a real-time bot-mitigation system for high-traffic product launches, combining behavioural analysis and session scoring to block automated requests with a 93% detection rate.
- AirMouse3D | Rust, Python, Firebase, Mobile Sensors | github.com/ravindran-dev/AirMouse3D**
- Created a 3D air-mouse application that translates hand gestures and accelerometer data into cursor movement, delivering 90% gesture-recognition precision across varied environments.
- AI Soft Skill Coach | Python, Agentic AI, LLMs, React, Firebase**
- Developed an AI-powered interview preparation tool that parses candidate resumes and generates personalised, role-specific mock-interview questions using large language models (LLMs). TGF 2.0 2026 Finalist — Top 10 out of 300 teams.

TECHNICAL SKILLS

Languages: Python, C, C++, Go, Rust, TypeScript, JavaScript, SQL, CUDA, Lua
AI and ML: Computer Vision, Deep Learning, NLP, Generative AI, LLMs, RAG, MLOps, GPU Optimization, Neural Networks, Object Detection, Model Deployment
Frameworks: PyTorch, TinyGrad, TensorRT, OpenCV, React.js, Streamlit, REST APIs
Platforms: Linux, Docker, Git, CI/CD, MQTT, Firebase, Redis, Nginx, AWS

ACHIEVEMENTS

- LeetCode Knight — peak contest rating 1,978 (top competitive programming tier)
- Ti Forge Hackathon 2026 — First place; awarded INR 50,000 cash prize
- AI for Sustainability Hackathon — First place; awarded USD 125 and Amazon vouchers worth 300 AED
- ICPC 2025 Preliminary Round — Global Rank 2,605 out of tens of thousands of participants

EDUCATION

B.E. Computer Science and Engineering (AI and ML) Sept 2024 – Present
Chennai Institute of Technology | CGPA: 9.24 / 10

CERTIFICATIONS

- NPTEL** - Internet of Things; 12-week course, 75% score
- Cisco** - CCNA, Cybersecurity, Python Fundamentals, and Modern AI (scores: 90–100%)
- Udemy Courses** - Full Stack Development.